DETAIL SPECIFICATION SHEET

PENETRATORS, MULTIPLE CABLE, COMPACT RECTANGULAR METAL FRAME AND BLOCKS, ELECTRICAL AND FIBER OPTIC CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-24705.

COMPACT RECTANGULAR CABLE PENETRATOR.

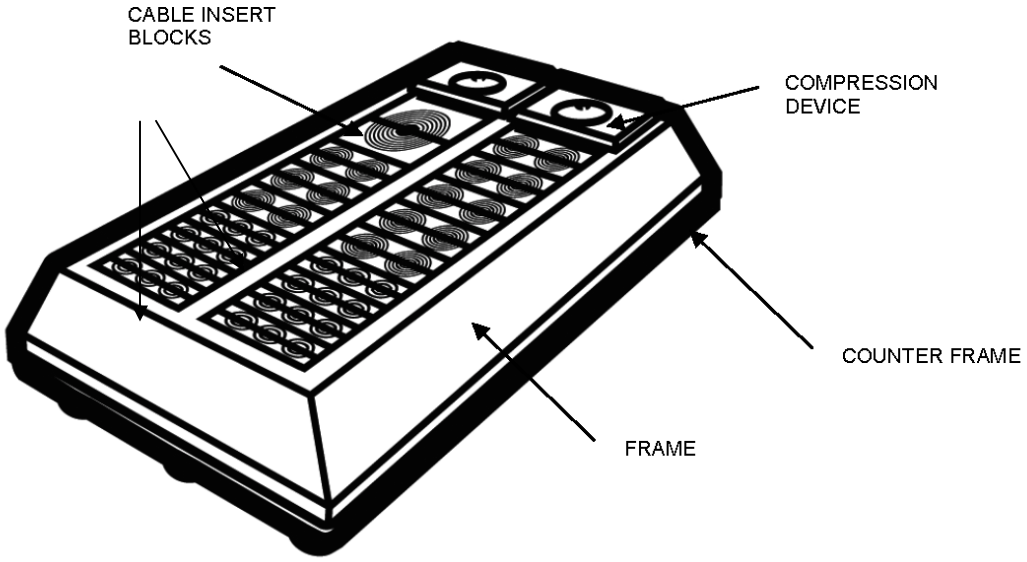
Intended use. Compact rectangular metal frame, cable insert blocks, and filler insert blocks are intended for use with fiber optic cables as specified in MIL-PRF-85045 and interconnection boxes as specified in MIL-DTL-24728/9, MIL-DTL-24728/10, electrical enclosures, and electronic equipment enclosures having cutouts of 215.9 mm ±2.5 mm (8.5 inches ±.1 inch) by 110 mm ±2.5 mm (4.33 inches ±.1 inch). The compact rectangular metal frame, cable insert blocks, and filler insert blocks specified by this specification sheet shall not be used to seal bulkheads or deck penetrations.

Compact rectangular cable penetrator assembly (figure 1). Components required for each compact rectangular cable penetrator specified by this specification sheet shall be as follows:

a. One compact rectangular metal frame assembly furnished with associated hardware (see figure 2).
   (1) Metal frame
   (2) Compression devices (quantity 2)
   (3) Counter frame
   (4) Hexagon nuts (quantity 10)
   (5) O-ring
   (6) Threaded inserts (quantity 10)

b. Compact cable insert blocks (see figures 3, 4, and 5) and filler insert blocks (see figure 6) to fill the interior of the compact rectangular metal frame assembly. Blocks are furnished separately.

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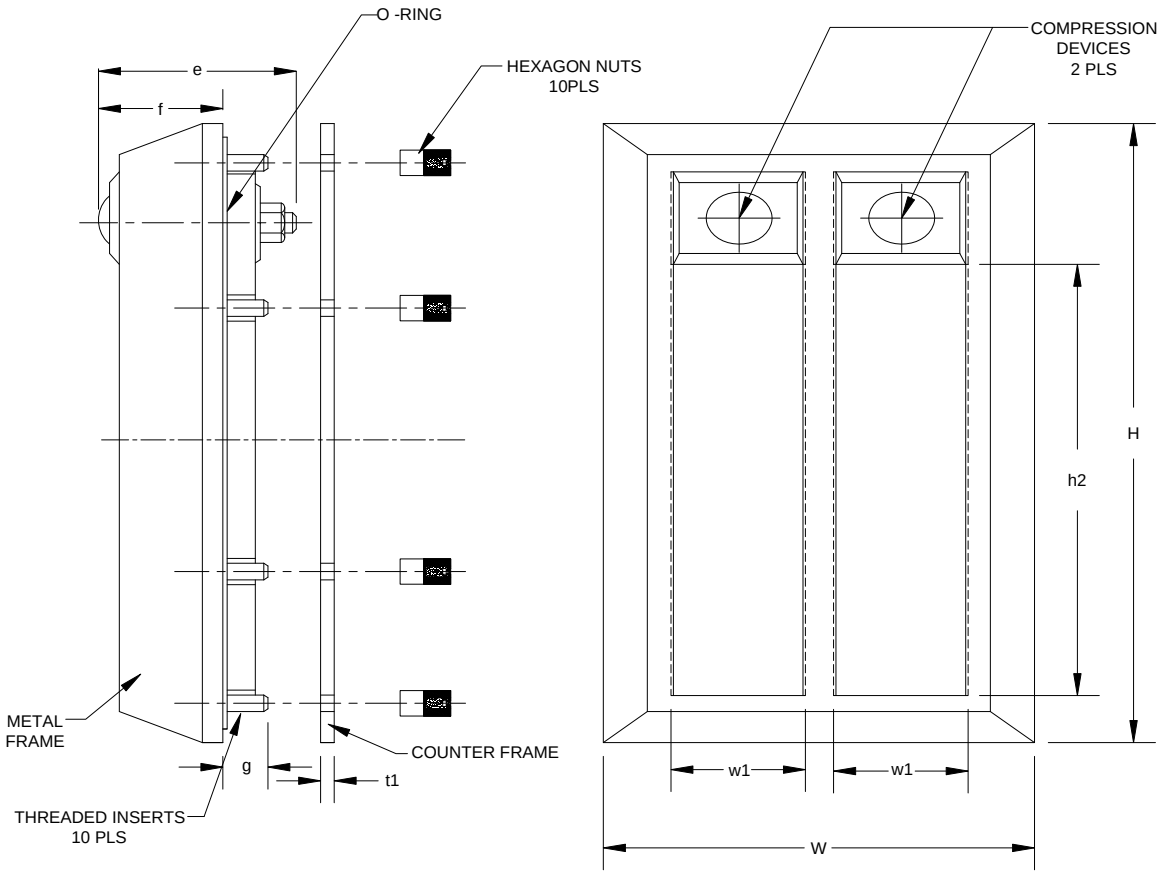
NOTES:

1. Drawing is not to scale.
2. Unless otherwise specified herein, tolerances for molded rubber parts shall be in accordance with the Rubber Manufacturers Association (RMA) Drawing Designation A3 Commercial tolerance specified in MIL-HDBK-149, in the table entitled “Standard dimensional tolerances - molded solid rubber products rubber manufacturers association (RMA) (69)”.

FIGURE 1. Compact rectangular cable penetrator - assembled view – typical.

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Compact rectangular frame and associated hardware. Compact rectangular frame and hardware shall be in accordance with figure 2.



Item type	Item material	Item designation	e	f	g	h2	t1	w1	W	H
F	A	01	59.0 (2.32)	38.0 (1.50)	1/	160.0 (6.30)	4.00 (.16)	40.0 (1.58)	130.0 (5.12)	230.0 (9.06)

NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Single decimal place tolerances are ±2.5 mm (±.098 inch).
5. Two decimal place tolerances are ±0.38 mm (±.015 inch).

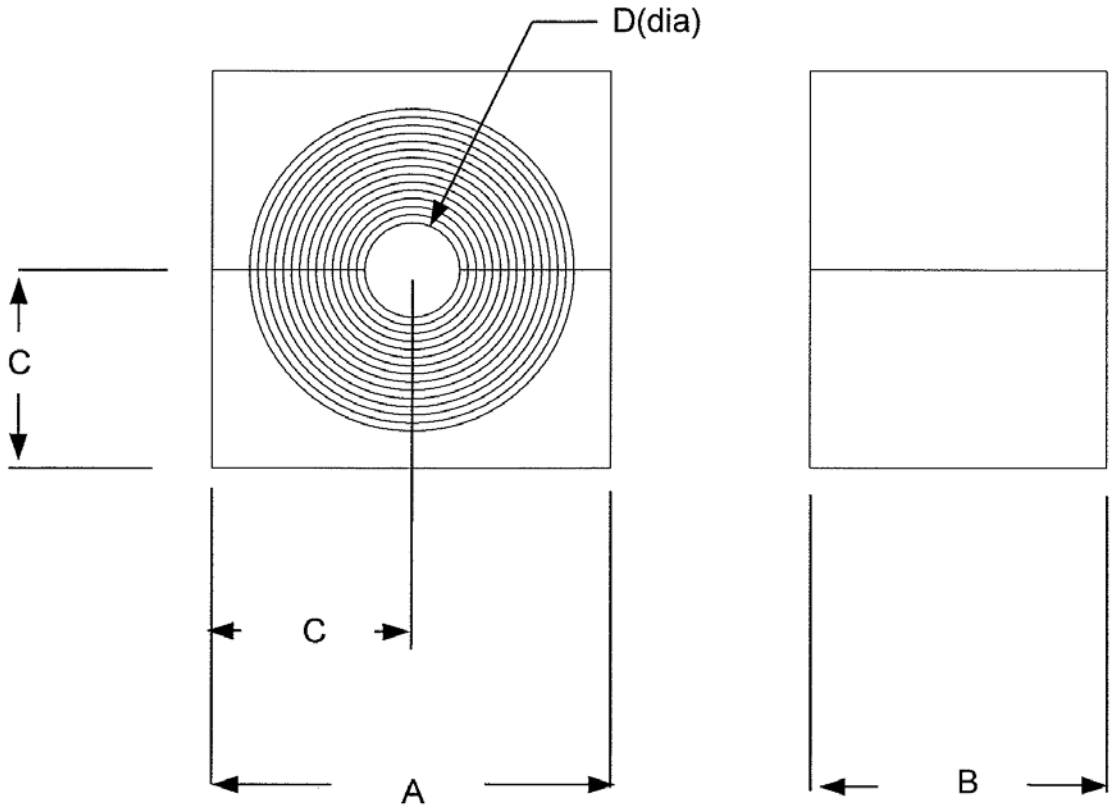
FOOTNOTES:

1/ Frame can be mounted in enclosures that have a panel thickness range of 3.0 – 7.0 mm (.12 - .28 inch).

FIGURE 2. Compact rectangular metal frame and associated hardware.

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Compact two-piece square-faced one cable insert block. Compact two-piece square-faced one cable insert blocks shall be in accordance with figure 3.



Item type	Item material	Item designation	A	B	C	D
B	N	40	40.0 (1.57)	30.0 (1.18)	20.0 (.79)	10.0 (.39)

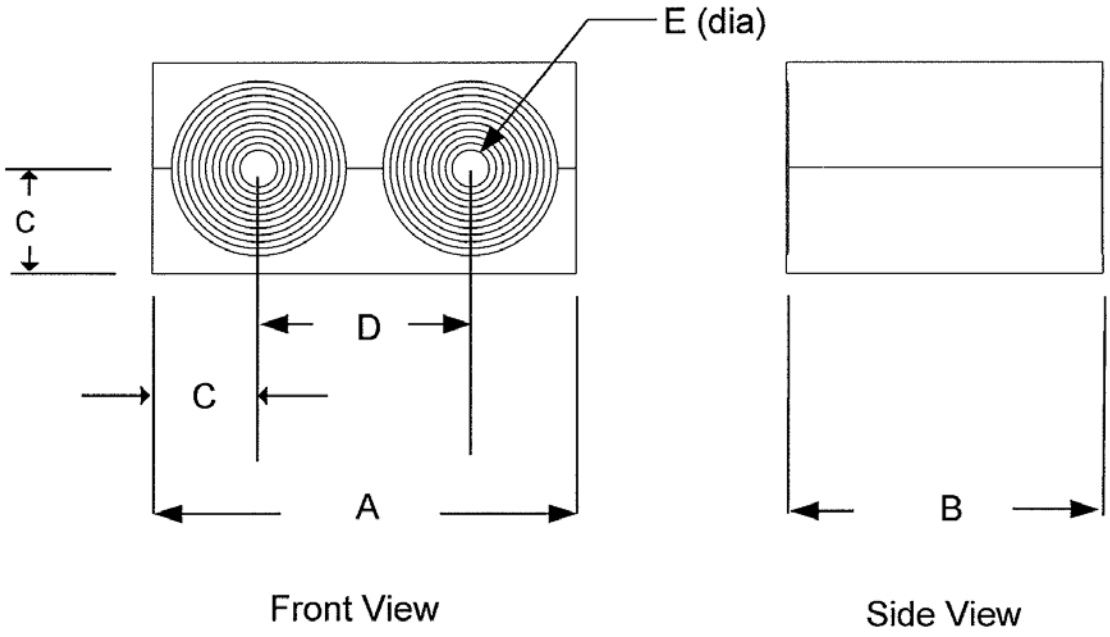
NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Item designation is an integer equal to two times the C dimension.

FIGURE 3. Compact two-piece squared-faced one cable insert block.

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Compact two-piece rectangular-faced two cable insert block. Compact two-piece rectangular-faced two cable insert blocks shall be in accordance with figure 4.



Item type	Item material	Item designation	A	B	C	D	E
B	N	20	40.0 (1.57)	30.0 (1.18)	10.0 (.39)	20.0 (.79)	3.2 (.13)

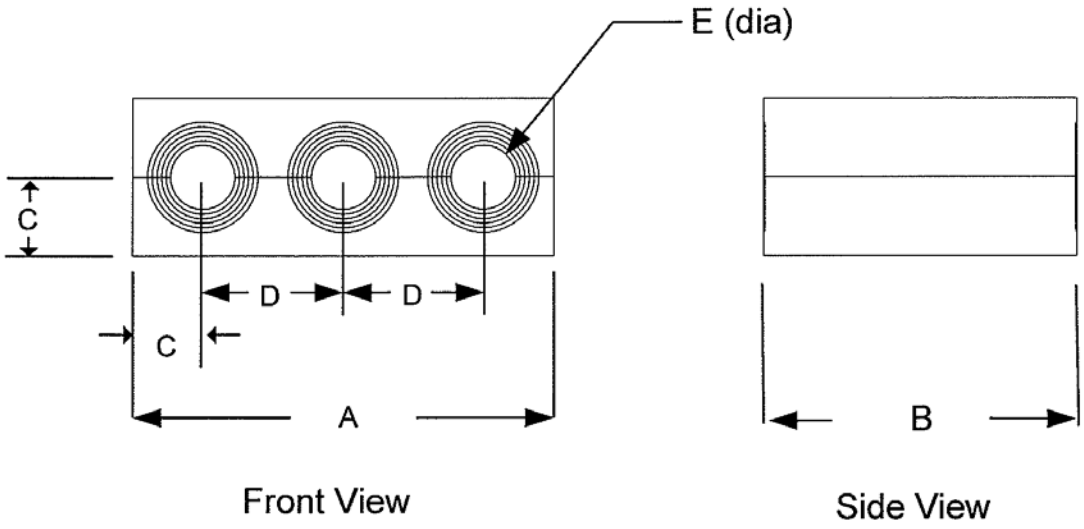
NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Item designation is an integer equal to two times the C dimension..

FIGURE 4. Compact two-piece rectangular-faced two-cable insert block.

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Compact two-piece rectangular-faced three cable insert block. Compact two-piece rectangular-faced three cable insert blocks shall be in accordance with figure 5.



Item type	Item material	Item designation	A	B	C	D	E
B	N	15	40.0 (1.57)	30.0 (1.18)	7.5 (.30)	12.5 (.49)	4.0 (.16)

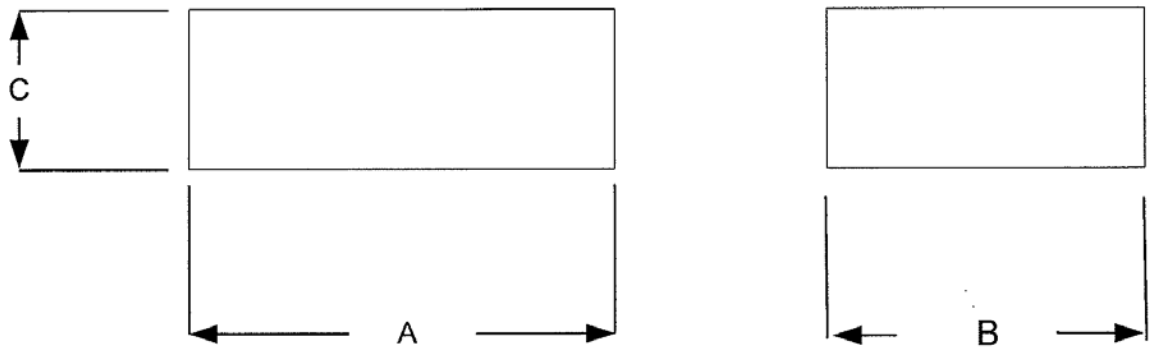
NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Item designation is an integer equal to two times the C dimension.

FIGURE 5. Compact two-piece rectangular-faced three cable insert block.

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Compact rectangular filler insert block. Compact rectangular filler insert blocks shall be in accordance with figure 6.



Item type	Item material	Item designation	A	B	C
I	N	05	40.0 (1.57)	30.0 (1.18)	5.0 (.20)
		10	40.0 (1.57)	30.0 (1.18)	10.0(.39)
		20	40.0 (1.57)	30.0 (1.18)	20.0 (.79)
		40	40.0 (1.57)	30.0 (1.18)	40.0 (1.57)

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Item designation is an integer equal to two times the C dimension.

FIGURE 6. Compact rectangular filler insert block.

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Part or identifying number (PIN) for compact rectangular cable penetrator assembly. Compact rectangular cable penetrator components are differentiated by an item designation in accordance with table I. An example of a PIN for a compact filler insert block 5 mm high of non-specified material is provided below (see MIL-DTL-24705 for complete details).

M	**24705**	**/4**	**-**	**I**	**N**	**05**
Prefix for Military Specification	Specification Number	Spec Sheet No.		Item Type (see table I)	Item Material (see table I)	Item Designation (see table I)

TABLE I. PIN.

Item type	Item material	Item designation	Description	Figure
F	A	01	Compact rectangular metal frame	2
B	N	40	Compact two-piece square-faced one cable insert block	3
B	N	20	Compact two-piece rectangular-faced two cable insert block	4
B	N	15	Compact two-piece rectangular-faced three cable insert block	5
I	N	05	Compact filler insert block	6
I	N	10	Compact filler insert block	6
I	N	20	Compact filler insert block	6
I	N	40	Compact filler insert block	6

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REQUIREMENTS.

Requirements for compact rectangular cable penetrators are as described in table II.

TABLE II. Requirements.

	MIL-DTL-24705/4 requirement
Material, alloys, aluminum	Applicable, with the following modification: Aluminum alloy in accordance with ASTM B179 shall be used for the compact rectangular metal frame and counterframe.
Material, corrosion-resisting steel (CRES)	Applicable, with the following modification: CRES 303 shall be used for the hexagon nuts. CRES 304 in accordance with ASTM A276 shall be used for the threaded inserts.
Construction, General	The cable insert blocks shall have peelable layers to accommodate different size cables and shall be in accordance with figures 1 through 6.
Threads	Applicable, with the following modification: Metric threads in accordance with FED-STD-H28/21and FED-STD-H28/22 shall be used. The nominal size and thread pitch shall be M6 x 1.
Vibration	Multiple cable penetrators shall show no evidence of broken, deformed, or displaced parts, chips, cracks, or physical damage when tested in accordance with 4.6.3 as modified herein.
High Impact (HI) Shock	Multiple cable penetrators shall show no evidence of broken, deformed, or displaced parts, chips, cracks, or physical damage when tested in accordance with 4.6.4 as modified herein.
Watertight	Not applicable.
Nonmagnetic	Multiple cable penetrators shall be nonmagnetic.
Electromagnetic compatibility (EMC)	Not applicable.
Salt spray	The compact rectangular cable penetrator assembly shall show no sign of salt penetration or corrosive effects when tested as specified in 4.6.8 as modified herein.
Fire stop	Not applicable.
Durometer	65 to 87 durometer
Compression	2 percent (maximum) percentage change when tested as specified in 4.6.11 as modified herein.
Expansion	1.5 percent (maximum).
Fluid immersion	The change in durometer hardness shall be not greater than ±5 percent when tested as specified in 4.6.13 and modified herein.
Infared (IR) signature	Not applicable.
Spraytight	The compact rectangular cable penetrator assembly shall meet the spraytight requirement specified in MIL-STD-108 when tested as specified herein.
Fungus	Materials used in manufacturing subcomponents shall be listed as fungus inert in guideline 4 of MIL-HDBK-454, or shall be tested as specified herein. Following the test, the polymeric material shall show sparse or very restricted microbial growth and reproduction with minor or inhibited substrate utilization not exceeding MIL-STD-810 Rating I for fungus growth. There shall be little or no chemical, physical, or structural change detectable.

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TABLE II. Requirements - Continued.

	MIL-DTL-24705/4 requirement
Marking: frames and accessories	The identification marking shall be: a. Vendor Part Number. b. Manufacturer's Name or Logo. c. Military Part or Identifying Number, in lieu of being included on item may be included on packaging.
Marking: cable and filler insert blocks	The identification marking shall be: a. Vendor Part Number. b. Manufacturer's Name or Logo. c. Military Part or Identifying Number, in lieu of marking on item it may be included on packaging.

VERIFICATION.

Verification methods required for compact rectangular cable penetrators are as described in table III.

TABLE III. Verification.

	MIL-DTL-24705/4 verification
Vibration	Testing may be performed with the compact rectangular cable penetrator assembly installed in a qualified MIL-DTL-24728/9 or MIL-DTL-24728/10 fiber optic interconnection box. Following the test, the multiple cable penetrator test specimen shall pass the shock test specified in 4.6.4 as modified herein; shall pass the spraytight test specified herein; and shall show no evidence of broken, deformed, or displaced parts, chips, cracks, or physical damage; and in that order.
Shock	Testing may be performed with the compact rectangular cable penetrator assembly installed in a qualified MIL-DTL-24728/9 or MIL-DTL-24728/10 fiber optic interconnection box. Following the test, the multiple cable penetrator shall pass the spraytight test specified herein.
Salt spray test	The multiple cable penetrator test specimen shall be tested in accordance with MIL-STD-810, Method 509. The total time (exposure time and drying time) shall be 96 hours in accordance with MIL-STD-810, Method 509. Testing may be performed with the compact rectangular cable penetrator assembly installed in a qualified MIL-DTL-24728/9 or MIL-DTL-24728/10 fiber optic interconnection box.
Compression	Each sample of insert material submitted for qualification shall be 29 mm in diameter x 10 mm. Samples shall be tested in accordance with ASTM D395, Method B, option 1.
Expansion	The sample size shall be one BN40 insert block.
Fluid immersion	Sample size shall be three BN40 blocks. After each fluid immersion test, the samples shall be removed from the test fluid and the insert material tested and shall pass test specified in 3.5.8.

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TABLE III. Verification - Continued.

	MIL-DTL-24705/4 verification
Spraytight	The compact rectangular cable penetrator assembly shall be tested in accordance with MIL-STD-108 for spraytight. Testing may be performed with the compact rectangular cable penetrator assembly installed in a qualified MIL-DTL-24728/9 or MIL-DTL-24728/10 fiber optic interconnection box.
Fungus	Materials not listed as fungus inert in guideline 4 of MIL-HDBK-454 shall be tested in accordance with TIA-455-56.

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Qualification inspection. Qualification inspection shall consist of the examination and tests specified herein on compact rectangular cable penetrator test specimens configured as shown on figure 7. The components for the test specimen shall be as specified on figure 7. Each component listed shall be required for each test specimen.

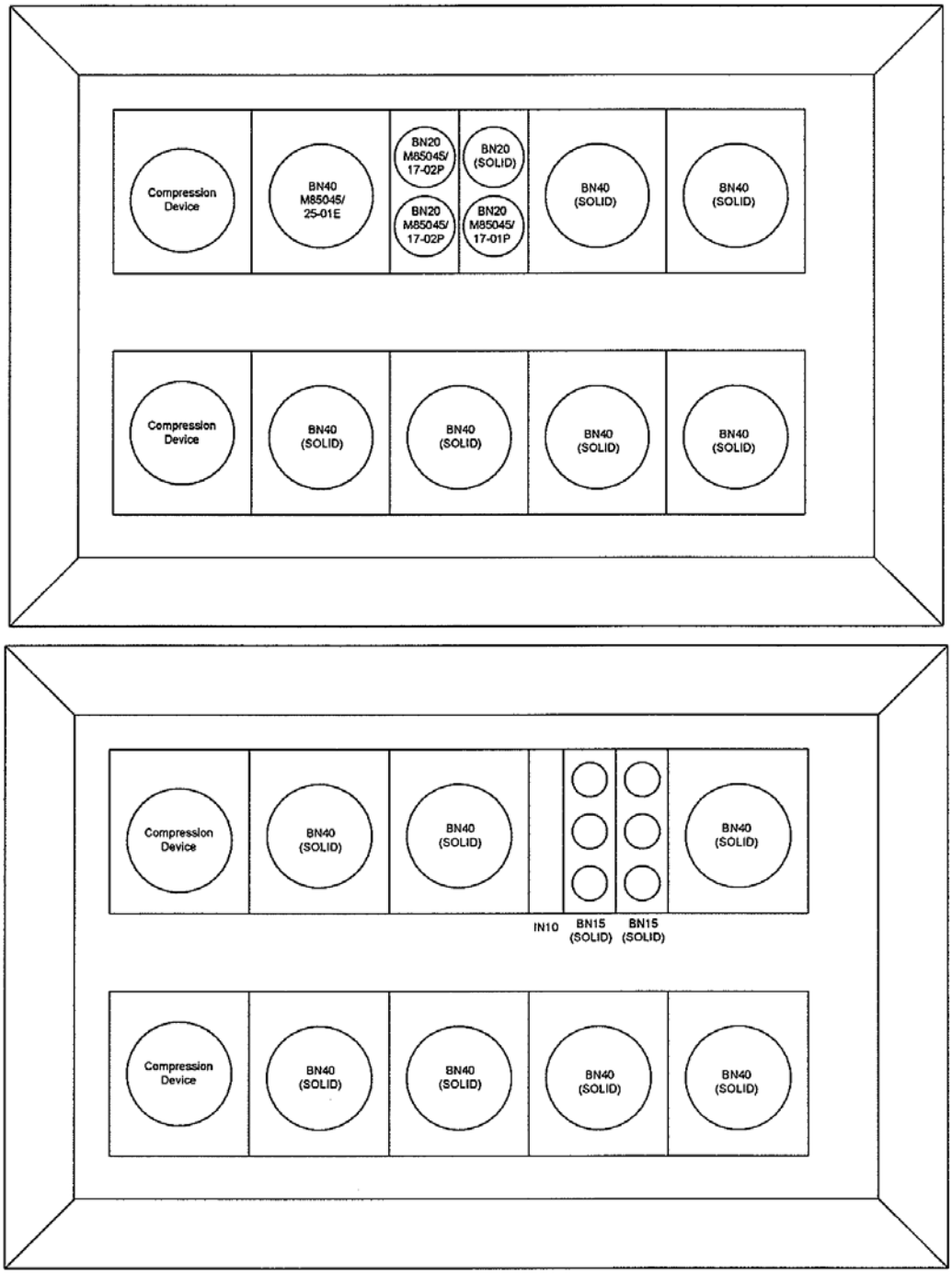


FIGURE 7. Qualification test specimen configurations 3/.

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Item	Quantity – part/description
Frame	2-FA01
Compact two-piece cable insert blocks	1-BN40 with specified test cables
	2-BN20 with specified test cables
Compact two-piece cable insert blocks 1/	13-BN40 2/
	2-BN15 2/
Filler insert blocks	1-IN10
Test cables	1-M85045/25-01E
	2-M85045/17-02P
	1-M85045/17-01P
Lubricant	As required – C-T-91

NOTES:

1/ The compact two-piece insert blocks can also be used as blanking insert blocks.

2/ Compact two-piece insert block with all layers and core intact.

3/ "(SOLID)" refers to the compact two-piece cable insert block with all layers and core intact.

FIGURE 7. Qualification test specimen configurations 3/ - Continued.

Qualification by similarity. Qualified manufacturers who produce cable insert blocks and filler blocks for MIL-DTL-24705/1 are qualified under this specification sheet if their product passes the size inspection, vibration, shock, and spraytight requirements specified herein provided the only difference between the previously qualified blocks and the blocks under test is the depth of the block identified as dimension B in figures 3, 4, 5, and 6.

Conformance inspection. Expansion shall be added to MIL-DTL-24705 Group B inspection.

REFERENCE DOCUMENTS.

General. The documents listed in this section are specified in this specification sheet.

Government documents (specifications, standards, and handbooks). The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28/21 - Screw-Thread Standards for Federal Services Section 21 Metric Screw-Threads

FED-STD-H28/22 - Screw-Thread Standards for Federal Services Section 22 Gages and Gaging for Metric Screw Threads-M and MJ Thread Forms

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-24728/9 - Interconnection Box, Cable Routing, Fiber Optic, Spraytight, 712 X 1372 mm

MIL-DTL-24728/10 - Interconnection Box, Cable Routing, Fiber Optic, Spraytight, 712 X 1017 mm

MIL-PRF-85045 - Cables, Fiber Optics, (Metric), General Specification for

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B179 - Standard Specification for Aluminum Alloys in Ingot and Molten Forms for Castings from All Casting Processes

(Copies of this document are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-455-56 - Test Method for Evaluating Fungus Resistance of Optical Fiber and Cable

(Copies of this document are available from the Telecommunications Industry Association, 2500 Wilson Blvd., Suite 300, Arlington, VA 22201-3834 or online at www.tiaonline.org.)

Custodians:
Navy – SH
DLA – CC

Preparing Activity:
Navy – SH
(Project 5975-2009-001)

Review Activities:
DLA – GS
GSA – FAS

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